HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2025)

Student No.: 17 Name: _____ Marks: / 50

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. If $(3c+1)(d-4) = 2d(5c-1)$, then $c =$

A. $\dfrac{3d-4}{7d+12}$.

B. $\dfrac{3d+4}{7d-12}$.

C. $\dfrac{7d-12}{3d+4}$.

D. $\dfrac{7d+12}{3d-4}$.

(DSE-2025 PII Q04)

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2. Let k be a constant. Solve the equation $x^2 + 4x = k^2 - 2k - 3$.

A. $x = k - 3$ or $x = -k - 1$

B. $x = k - 3$ or $x = -k + 1$

C. $x = k + 3$ or $x = -k - 1$

D. $x = k + 3$ or $x = -k + 1$

(DSE-2025 PII Q05)

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3. If $x = 5.67$ (correct to 2 decimal places), find the range of values of x.

A. $5.66 < x \leq 5.68$

B. $5.66 \leq x < 5.68$

C. $5.665 < x \leq 5.675$

D. $5.665 \leq x < 5.675$

(DSE-2025 PII Q06)

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4. If α, β and γ are non-zero constants such that $(\alpha+2\beta):(\beta+2\gamma):(\gamma+2\alpha)=4:9:5$, then $\alpha:\beta=$

A. $2:5$.

B. $5:2$.

C. $128:149$.

D. $149:128$.

(DSE-2025 PII Q11)

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5. It is given that z varies directly as the cube of x and inversely as the square of y. When $x = 3$ and $y = 6$, $z = 3$. When $x = 5$ and $y = 2$, $z =$

A. 5.

B. 25.

C. 125.

D. 243.

(DSE-2025 PII Q12)

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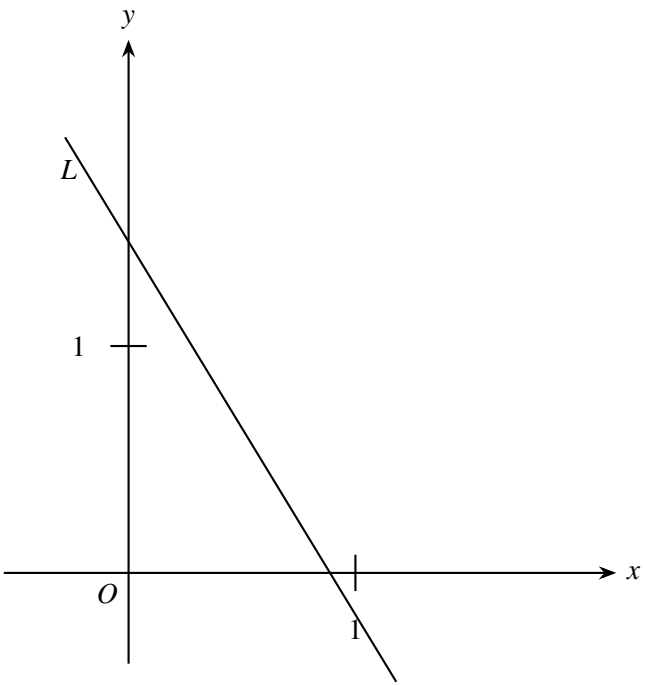
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6. In the figure, the equation of the straight line L is $px + qy = 7$. Which of the following is/are true?

I. $p > 7$

II. $q > 7$

III. $q > p$



A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2025 PII Q14)

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7. The perimeter of the sector OMN is 12π cm, where O is the centre of the sector OMN. If the radius of the sector OMN is 3π cm, which of the following are true?

I. The area of the sector OMN is $9\pi^2$ cm^2.

II. The perimeter of $\triangle OMN$ is less than 35 cm.

III. The angle of the sector OMN is greater than $100°$.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2025 PII Q15)

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8. The height and the total surface area of a solid right circular cylinder are 35 cm and 492π cm^2 respectively. If the radius of a solid sphere is equal to the base radius of the circular cylinder, then the volume of the sphere is

A. 144π cm^3.

B. 288π cm^3.

C. 576π cm^3.

D. 864π cm^3.

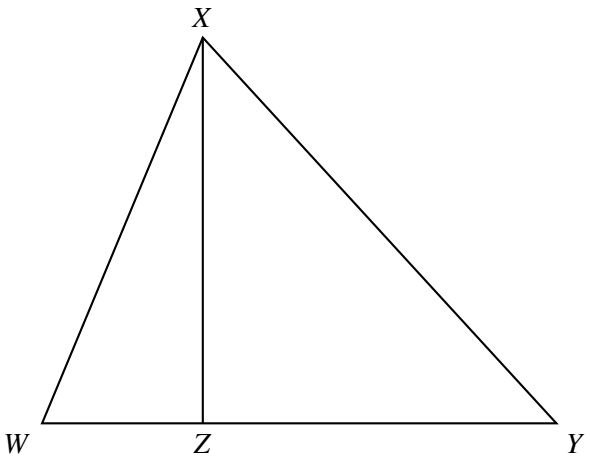
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9. The figure shows the triangle WXY. Let Z be a point lying on WY such that $WY : XY = XY : YZ$. If $WX = 65\,\text{cm}$, $WZ = 25\,\text{cm}$ and $XZ = 60\,\text{cm}$, then $XY =$



A. 131 cm.

B. 144 cm.

C. 156 cm.

D. 169 cm.

(DSE-2025 PII Q18)

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10. It is given that $PQRS$ is a trapezium with $PQ \,//\, SR$. If $PS = 41\,\text{cm}$, $RS = 53\,\text{cm}$, $\angle PSR = 120^\circ$ and $\angle QRS = 150^\circ$, then $PQ =$

A. 82 cm.

B. 100 cm.

C. 106 cm.

D. 135 cm.

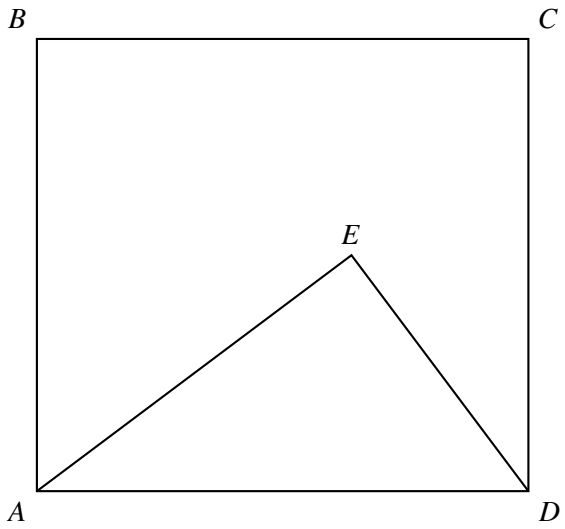
(DSE-2025 PII Q20)

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11. In the figure, $ABCD$ is a rectangle. It is given that $AE = 20\,\text{cm}$ and $\angle AED = 90°$. If the area of $\triangle ADE$ is $150\,\text{cm}^2$, then the perpendicular distance from E to CD is



A. 8 cm.

B. 9 cm.

C. 12 cm.

D. 15 cm.

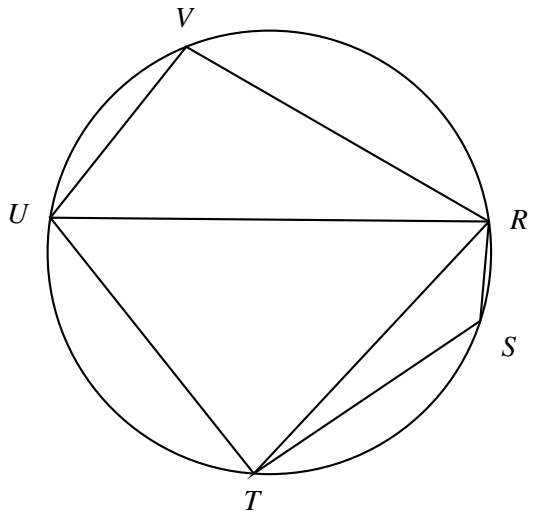
(DSE-2025 PII Q21)

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12. In the figure, $RSTUV$ is a circle. It is given that $RT \,//\, VU$ and RT is the angle bisector of $\angle SRU$. If $\angle RTS = \angle URV = 33°$, then $\angle RUT =$



A. 36°.

B. 49°.

C. 65°.

D. 71°.

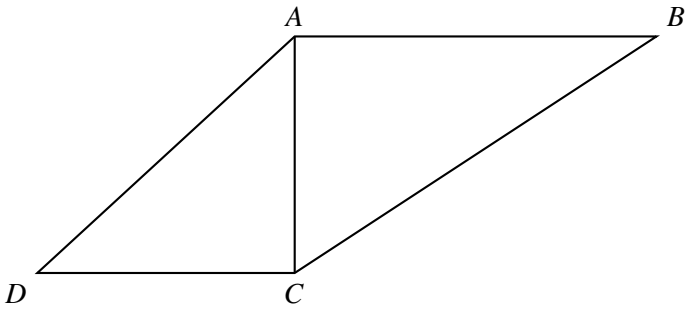
(DSE-2025 PII Q22)

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13. The figure shows the quadrilateral $ABCD$. If $\angle ABC + \angle ADC = \angle ACD = \angle BAC = 90°$, which of the following must be true?



A. $\tan\angle ACB = \dfrac{AB}{AD}$

B. $\tan\angle ACB = \dfrac{AB}{CD}$

C. $\tan\angle ADC = \dfrac{BC}{AD}$

D. $\tan\angle ADC = \dfrac{BC}{CD}$

(DSE-2025 PII Q23)

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14. The polar coordinates of the points X, Y and Z are $(1, 20^\circ)$, $(2, 80^\circ)$ and (r, θ) respectively, where $20^\circ < \theta < 80^\circ$. If $\triangle XYZ$ is an equilateral triangle, find r.

A. $\sqrt{3}$

B. $\sqrt{5}$

C. $\sqrt{7}$

D. $\sqrt{10}$

(DSE-2025 PII Q24)

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15. The straight lines L_1 and L_2 are perpendicular to each other. The equations of L_1 and L_2 are $3x+4y-20=0$ and $mx+ny-20=0$ respectively, where m and n are constants. Suppose that L_1 and L_2 cut the x-axis at the points A and B respectively. Let C be the point of intersection of L_1 and L_2. It is given that C lies above the x-axis. If the area of $\triangle ABC$ is 6, find n.

A. -12

B. -9

C. 9

D. 12

(DSE-2025 PII Q26)

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16. The coordinates of the centre of the circle C are $(7,-5)$. If C cuts the x-axis at the points P and Q such that $PQ = 24$, then the equation of C is

A. $x^2+y^2-14x+10y-95=0$.

B. $x^2+y^2+14x-10y-95=0$.

C. $x^2+y^2-14x+10y-119=0$.

D. $x^2+y^2+14x-10y-119=0$.

(DSE-2025 PII Q27)

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17. A box contains six cards numbered 2, 2, 2, 3, 4 and 5 respectively. In a game, a number is randomly drawn from the box and a certain number of tokens will be got according to the following table:

Number drawn	2	3	4	5
Number of tokens got	10	15	25	50

Find the expected number of tokens got in the game.

A. 3

B. 4

C. 20

D. 25

(DSE-2025 PII Q28)

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18. The L.C.M. of p^2-4q^2, p^3-8q^3 and $(p+2q)(p^2-4q^2)$ is

A. $p-2q$.

B. p^2-4q^2.

C. $(p+2q)^2(p^3-8q^3)$.

D. $(p+2q)(p^2-4q^2)(p^3-8q^3)$.

(DSE-2025 PII Q32)

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19. On the same rectangular coordinate system, the graph of $y = \log_a x$ cuts the graph of $y = a^x$ and the x-axis at the points P and Q respectively, where a is a positive constant. Denote the origin by O. Which of the following are true?

I. $a < 1$

II. $OQ > a$

III. $\angle POQ = 45°$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2025 PII Q34)

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20. Let p, q and r be non-zero real numbers. If p, q, r is an arithmetic sequence, which of the following must be true?

I. 3^p, 3^q, 3^r is a geometric sequence.

II. $\frac{5}{p}$, $\frac{5}{q}$, $\frac{5}{r}$ is a geometric sequence.

III. $p-q$, $q-r$, $r-p$ is an arithmetic sequence.

A. I only

B. II only

C. I and III only

D. II and III only

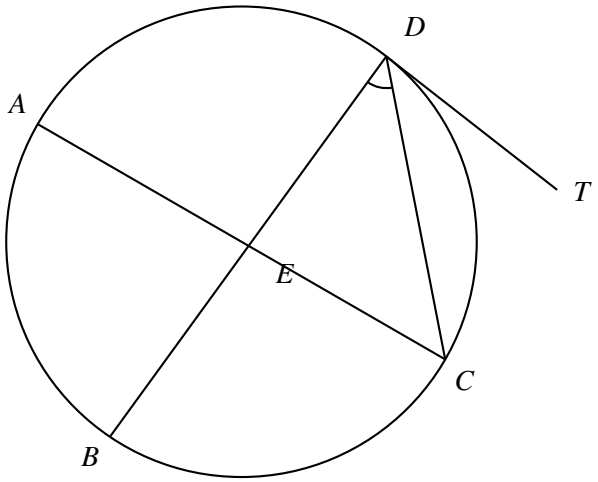
(DSE-2025 PII Q37)

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21. In the figure, AC is a diameter of the circle $ABCD$. Denote the point of intersection of AC and BD by E. It is given that TD is the tangent to the circle at D. If $\angle BEC = 96^\circ$ and $\angle CDT = 41^\circ$, then $\angle CDE =$



A. 47°.

B. 48°.

C. 52°.

D. 55°.

(DSE-2025 PII Q38)

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22. For $90° < \theta < 270°$, how many roots does the equation $\tan^3\theta = 2\tan\theta$ have?

A. 2

B. 3

C. 4

D. 5

(DSE-2025 PII Q39)

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23. A committee is formed by 2 managers, 4 officers and 12 clerks. If 7 members are selected from the committee to form a working group consisting of at least 1 manager, how many different working groups can be formed?

A. 16016

B. 20384

C. 22880

D. 31824

(DSE-2025 PII Q42)

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24. There are 9 cans of apple juice and 4 cans of grape juice in a bag. If 6 cans are randomly chosen from the bag at the same time, find the probability that at most 3 cans of grape juice are chosen.

A. $\frac{9}{13}$

B. $\frac{133}{143}$

C. $\frac{140}{143}$

D. $\frac{714}{715}$

(DSE-2025 PII Q43)

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25. The standard scores of a boy and a girl in a Mathematics test are -2 and z respectively. The standard deviation of the scores of the Mathematics test is 2 marks. If the difference of the test score of the boy and the test score of the girl is 6 marks, find z.

A. -5 or 1

B. -5 or 3

C. -3 or 1

D. -3 or 3

(DSE-2025 PII Q44)

I can finish _____ questions

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